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4110112 Rev A

Aurum™ Plasmid Midi Kit

Instruction Manual

For technical service, call your local Bio-Rad office, or in the US, call 1-800-4BIORAD (1-800-424-6723)

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Section 1
Introduction

The Aurum plasmid midi kit employs a special lysate filtration column to clarify the bacterial lysate, thus eliminating the requirement for a high-speed centrifugation step. This significantly reduces handling and increases the speed and convenience of large-scale plasmid purification. The lysate filtration column can be used in either a vacuum or spin format.

In the vacuum format, the lysate filtration midi column (yellow column) is attached in tandem to the plasmid midi column (green column) with a special adaptor cap, forming a "binary" column unit. This binary configuration allows plasmid binding and lysate filtration (clarification) to occur simultaneously, for a particularly rapid procedure. In the spin format, the lysate filtration midi column is placed into a standard 15 ml centrifuge tube, and lysate clarification is achieved by a short, low-speed centrifugation. The clarified lysate is then transferred into the plasmid midi column where binding, washing and elution are also achieved via brief, low-speed centrifugations. No subsequent alcohol precipitation is required.

The Aurum plasmid midi kit is optimized for the purification of up to 100 µg of plasmid DNA, rapidly and inexpensively, without the use of toxic reagents, high-speed centrifugations, or alcohol precipitations. Plasmid DNA purified with the Aurum plasmid midi kit is eluted into a small volume of aqueous buffer and is free of salts, bacterial chromosomal DNA, and RNA. The exceptional purity of the plasmid produced with this method is ideal for the most demanding applications, such as mammalian cell transfection.

Section 2
Kit Components

The Aurum plasmid midi kit contains the following components:

resuspension solution	20 ml
filtration lysis solution	25 ml
neutralization solution	40 ml
wash solution	25 ml (5x concentrate)
elution solution	16 ml
plasmid midi columns (green)	20
lysate filtration midi columns (yellow)	20
midi adaptor caps	20

The solutions are specifically formulated for the Aurum plasmid midi kit. They are NOT interchangeable with solutions used in other kits or protocols.

Section 3
Storage Conditions

Solutions, columns, and caps should be stored at room temperature. If precipitation is observed in any solution, warm solution to 37ºC to redissolve, and allow to return to room temperature before use. Do not expose any of the solutions to temperatures above 37ºC. If the kit is used infrequently, storage of the resuspension solution at 4ºC is recommended to preserve the RNase.

Section 4
Necessary Supplies

Equipment and supplies to be provided by the customer:

- High-speed centrifuge (fixed angle rotor) to pellet bacteria
- Low-speed (tabletop) centrifuge with swinging bucket rotor (recommended)
- 30–50 ml high-speed centrifuge tubes (Oak Ridge type)
- 15 ml conical low-speed centrifuge tubes
- 95–100% ethanol, or reagent-grade (denatured) ethanol 100 ml

Additional equipment required for vacuum format:

- Bio-Rad Aurum vacuum manifold with vacuum regulator and column adaptor plate (Cat. #732-6470) or other vacuum manifold with luer fittings (Figure 1).

Note: Please read Section 7, **Instrument Setup and Use for the Column Adaptor Plate** in the Aurum vacuum manifiold instruction manual for proper vacuum setup conditions.

- Vacuum source (capability of -20 to -23" Hg required)

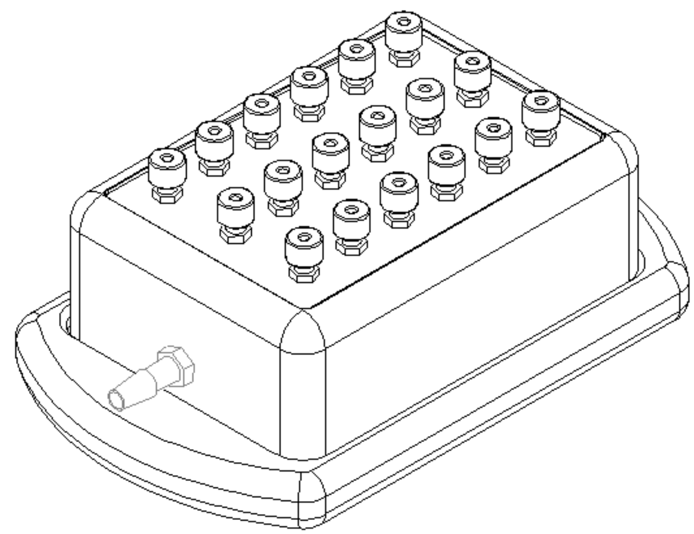


Fig. 1. Aurum vacuum manifold

Section 5
Guidelines for Using the Aurum Plasmid Midi Kit

Please read the following guidelines before starting the plasmid purification.

Bacterial Growth Guidelines:

- The Aurum plasmid midi kit can process bacteria grown in a variety of different broths, such as LB (Luria-Bertani broth), LBG (LB + 2% glycerol), SB (Super Broth) and 2x YT. For optimum performance, LB or LBG are recommended for most strains of *E. coli.* TB (Terrific Broth) cultures generally produce lower plasmid yields of more variable quality and are therefore not recommended.
- For optimum plasmid quality, plasmid propagation in an *endA*$^-$ host, such as DH5α, JM109, or XL1-Blue, is recommended.
- Spectrophotometric determination of the bacterial culture density is a REQUIREMENT for optimum performance.

  To determine the density of of a bacterial culture (OD_{600}), combine 50 µl of bacterial culture with 950 µl growth medium (1:20 dilution). Use the growth medium as a blank and take the spectrophotometric reading at λ = 600 nm. Multiply this figure by 20 to calculate the bacterial concentration. Depending upon the OD_{600} value, a specific volume of the culture will be selected to provide an optimum amount of bacteria for processing. To calculate the volume of bacterial culture required, use the following equation:

  (OD_{600} of undiluted culture*) x (culture volume in ml) = value in OD•ml

  For example, 50 OD•ml of bacteria would require 10 ml of an undiluted culture with an OD_{600} = 5.
- The protocol is designed to process 20–50 OD•ml of bacterial host. When processing 20–30 OD•ml, reduce the volumes of the resuspension, filtration lysis and neutralization solutions to 0.6 ml, 0.6, ml and 1.2 ml, respectively. This establishes an optimal ratio of reagent : bacterial cellular debris, which enhances the adhesion and removal of the cellular debris by the filtration column. Processing the maximum density of bacterial host (50 OD•ml) is recommended in order to maximize the concentration of plasmid in the eluate. However, do not process more than 50 OD•ml or less than 20 OD•ml.

*1 OD_{600} is equivalent to approximately 8 x 10^8 cells/ml

Filtration Guidelines:

- The filtration lysis solution is specifically formulated to allow lysate clarification via filtration. It is NOT interchangeable with alkaline solutions used in other kits or protocols.

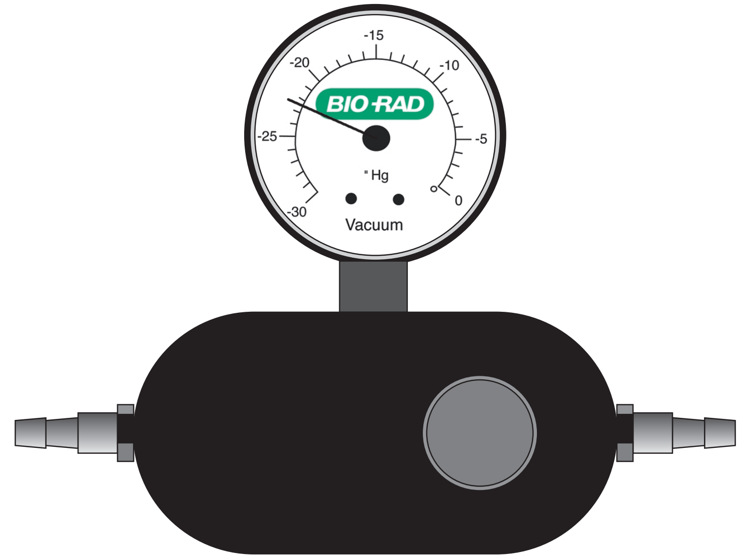


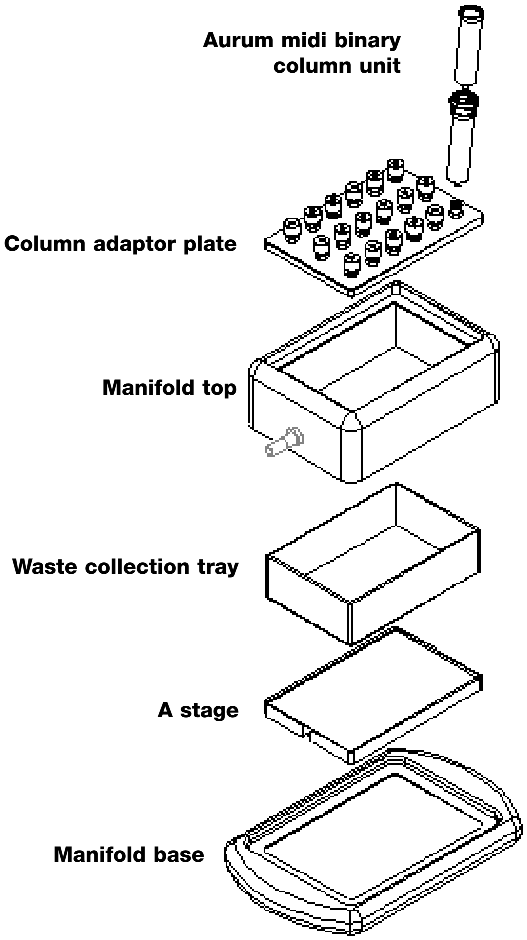
Fig. 2. Vacuum regulator with clear representation of the gauge

Vacuum Guidelines:

- The recommended operating range is -20 to -23 inches of mercury (" Hg). Do not exceed -25" Hg when performing this protocol. A vacuum regulator is **required** to establish the appropriate negative pressure (Figure 2).

Table 1. Pressure unit conversions

To convert from inches of mercury (" Hg) to:	Multiply by:
millimeters of mercury or torr (mm Hg, torr)	25.4
millibar (mbar)	33.85
atmospheres (atm)	0.03342
pounds per square inch (psi)	0.4912
kilopascals (kPa)	3.385



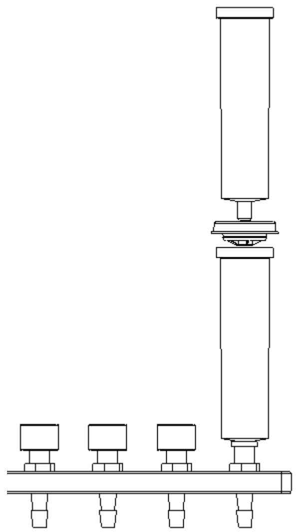


Fig. 3a. Vacuum setup conditions for the Aurum plasmid midi kit

Fig. 3b. Connection of Aurum midi binary column unit to column adaptor plate

- If negative pressure remains within the manifold and columns, the membranes and retainer ring may dislocate from the base of the green plasmid midi column when the adaptor cap is removed. The negative pressure MUST be reduced to "0" before removing the adaptor cap. If membrane dislocation should occur, use a clean blunt object or gloved finger to push the retainer and membranes back into position. This will not affect either the yield or quality of the plasmid preparation.

Centrifugation and Elution Guidelines:

- After centrifugation, if residual liquid remains above the membrane stack, centrifuge at 2,500 x g for an additional 5 min.
- To increase the concentration of plasmid in the eluate, the volume of elution solution may be reduced to 0.6 ml. Further reduction in the volume of the elution solution may result in reduced plasmid recovery from the column membranes.

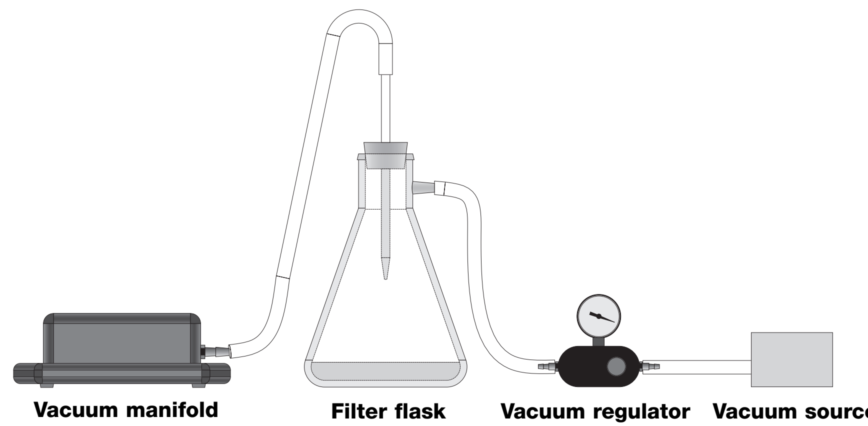


Fig. 4. Vacuum setup conditions

Section 6
Protocol

Vacuum Format

This procedure requires the Bio-Rad Aurum vacuum manifold and column adaptor plate (Cat. #732-6470), or any vacuum manifold with luer fittings. For proper vacuum setup conditions, please read **Instrument Setup and Use for the Column Adaptor Plate**, in the Aurum vacuum manifold instruction manual and see Figure 4. Please read the previous section "Guidelines for Using the Aurum Plasmid Midi Kit" before proceeding.

1. Transfer up to 50 OD•ml of an overnight culture to one or more 30–50 ml high-speed (Oak Ridge type) centrifuge tubes. Centrifuge the cells for 10 min at 4,000 x g. Alternatively, the cells can be pelleted in conical centrifuge tubes at a lower speed/g-force. After centrifugation, check that the bacterial pellet is firmly compacted against the bottom of the tube.

2. Invert the tube and pour off the supernatant. Blot the end of the tube against several layers of paper toweling.

3. Add 1 ml of resuspension solution to the bacterial pellet. Vortex or pipet up and down until the pellet is completely resuspended.

   Note: If processing ≤ 30 OD•ml of bacterial culture, reduce the volume of resuspension solution used to 0.6 ml.

4. Add 1 ml of filtration lysis solution and mix by inverting the tube briskly 6–8 times. DO NOT VORTEX. The solution will become viscous.

   Note: If processing ≤ 30 OD•ml of bacterial culture, reduce the volume of filtration lysis solution used to 0.6 ml.

   Note: The neutralization solution should be added to the lysate within 5 min of cell lysis.

5. Add 2 ml of neutralization solution and mix by inverting the tube briskly 6–8 times. DO NOT VORTEX. Allow the lysate to stand for 5 minutes before proceeding to the next step. The bacterial chromosomal DNA and cellular debris will form a visible white precipitate.

   Note: If processing ≤ 30 OD_{600} of bacterial culture, reduce the volume of neutralization solution used to 1.2 ml.

6. Remove the manifold top and place the A stage in the depression in the center of the manifold base. The etched letter A should be facing up.

7. Place the waste collection tray on top of the A stage. Replace the manifold top on the base, ensuring complete and uniform contact between the manifold top and base.

8. With the vacuum source off and the vacuum regulator completely open, set up the lysate filtration and plasmid midi columns in the following manner (refer to Figures 3 and 4):

   a. Position the column adaptor plate into the recess on the top of the manifold. The adaptor plate will seal itself against the manifold top when vacuum is applied.

   b. Place an adaptor cap into the top of each green plasmid midi column. Make sure that the cap is firmly seated on the column.

   c. Insert the luer end of a yellow lysate filtration midi column into the corresponding orifice in the adaptor cap. Push the column into place until a firm fit is established.

   d. Transfer this assembly to the column adaptor plate by pushing the luer end of the green plasmid midi column into one of the luer fittings on the column adaptor plate. Make sure that the plasmid midi column is firmly seated in the adaptor plate fitting.

   e. Close all unused luer fittings with luer caps.

9. With the vacuum turned off, pour the neutralized lysate from step 5 into the yellow lysate filtration column. Turn the vacuum on and close the vacuum regulator to increase the negative pressure to -20 to -23" Hg.

10. Continue to apply vacuum until no further filtrate is discharged from the yellow lysate filtration midi column, and no filtrate remains above the membrane stack at the base of the green plasmid midi column. Open the vacuum regulator until the vacuum gauge indicates 0" Hg.

    Note: Some droplets of filtrate may have been deposited on the wall of the green column. This will not affect purification.

11. The wash solution is supplied as a 5x concentrate. Add 4 volumes (100 ml) of 95–100% ethanol or reagent-grade (denatured) ethanol before initial use.

12. Remove and discard the lysate filtration midi column and adaptor cap from the plasmid midi column. Add 5 ml wash solution to the plasmid midi column, and close the vacuum regulator to increase the negative pressure to approximately -20 to -23" Hg. After all wash solution has

passed through the column, continue the vacuum for an additional 4 min to purge the column of residual wash solution. Open the vacuum regulator until the vacuum gauge indicates 0" Hg. Ensure that no droplets of wash solution remain near the top of the column.

Note: Do not allow the purge to continue for longer than 5 min or excessive drying of the column membranes may occur.

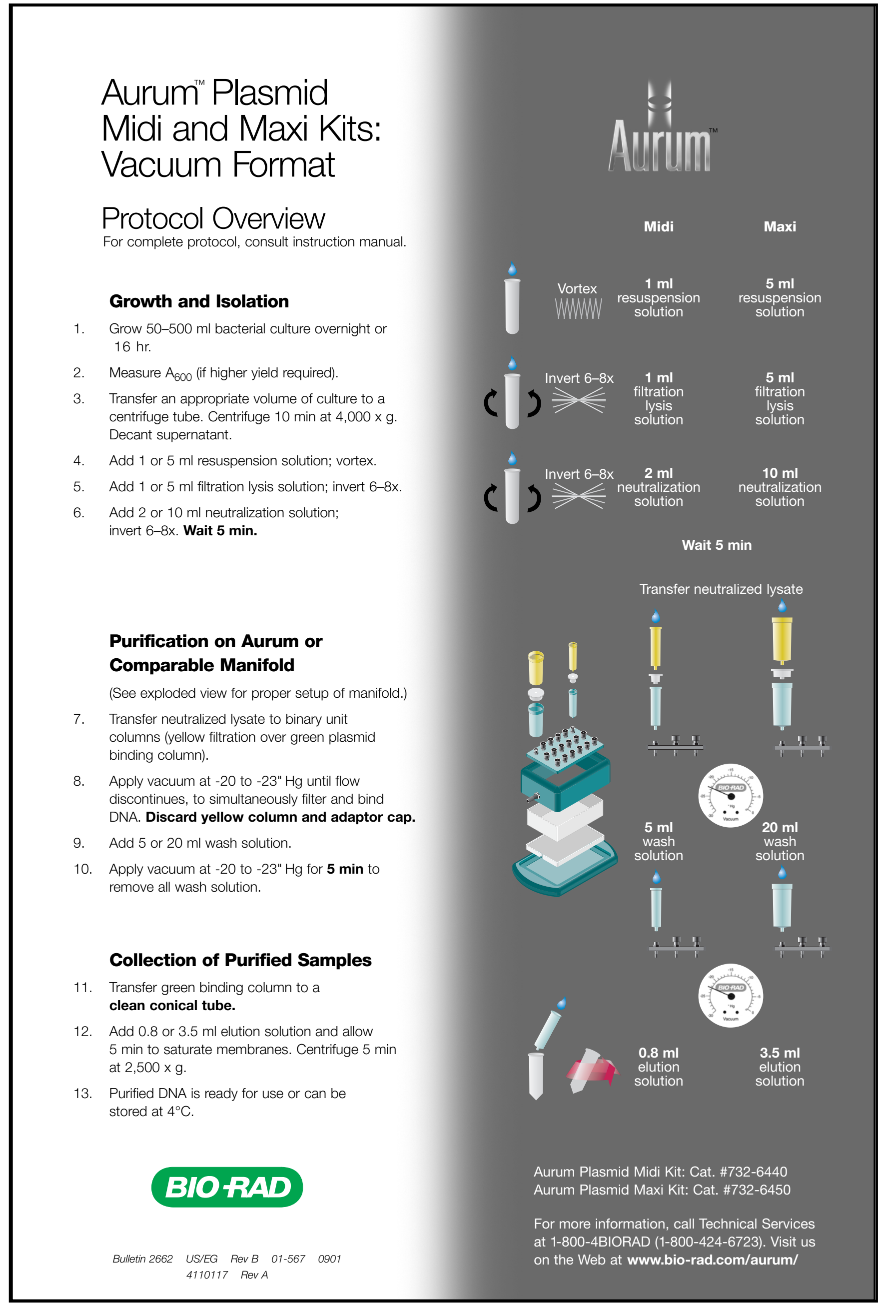
13. Remove the plasmid midi column from the vacuum manifold and transfer to a clean 15 ml conical tube. Add 0.8 ml elution solution directly onto the membranes at the base of the column and allow 5 min for the solution to saturate the membranes.

    Note: To obtain more concentrated plasmid DNA samples, the volume of elution solution may be reduced to 0.6 ml. Eluting with less than 0.6 ml may result in reduced plasmid yield.

14. Elute the plasmid DNA by centrifuging for 5 min at 2,500 x g.

    The eluted plasmid samples in the 15 ml tubes can be used immediately in mammalian cell transfections or in any other application. Alternatively, the plasmid can be stored at 4°C for later use.

    A protocol overview is available (see Figure 5).



Aurum™ Plasmid Midi and Maxi Kits: Vacuum Format

Protocol Overview

For complete protocol, consult instruction manual.

Growth and Isolation

1. Grow 50–500 ml bacterial culture overnight or 16 hr.
2. Measure A_{600} (if higher yield required).
3. Transfer an appropriate volume of culture to a centrifuge tube. Centrifuge 10 min at 4,000 x g. Decant supernatant.
4. Add 1 or 5 ml resuspension solution; vortex.
5. Add 1 or 5 ml filtration lysis solution; invert 6–8x.
6. Add 2 or 10 ml neutralization solution; invert 6–8x. **Wait 5 min.**

Purification on Aurum or Comparable Manifold

(See exploded view for proper setup of manifold.)

7. Transfer neutralized lysate to binary unit columns (yellow filtration over green plasmid binding column).
8. Apply vacuum at -20 to -23" Hg until flow discontinues, to simultaneously filter and bind DNA. **Discard yellow column and adaptor cap.**
9. Add 5 or 20 ml wash solution.
10. Apply vacuum at -20 to -23" Hg for **5 min** to remove all wash solution.

Collection of Purified Samples

11. Transfer green binding column to a **clean conical tube.**
12. Add 0.8 or 3.5 ml elution solution and allow 5 min to saturate membranes. Centrifuge 5 min at 2,500 x g.
13. Purified DNA is ready for use or can be stored at 4°C.

BIO-RAD

Aurum Plasmid Midi Kit: Cat. #732-6440
Aurum Plasmid Maxi Kit: Cat. #732-6450

For more information, call Technical Services at 1-800-4BIORAD (1-800-424-6723). Visit us on the Web at **www.bio-rad.com/aurum/**

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Fig. 5. Aurum plasmid midi protocol overview: vacuum format

Spin Format

This protocol should be followed if a vacuum manifold is unavailable. The enclosed adaptor caps will not be used. Please read the previous section "Guidelines for Using the Aurum Plasmid Midi Kit" before proceeding.

1. Transfer up to 50 OD•ml of an overnight culture to one or more 30–50 ml high-speed (Oak Ridge type) centrifuge tubes. Centrifuge the cells for 10 min at 4,000 x g. Alternatively, the cells can be pelleted in conical centrifuge tubes at a lower speed/g-force. After centrifugation, check that the bacterial pellet is firmly impacted against the bottom of the tube.

2. Invert the tube and pour off the supernatant. Blot the end of the tube against several layers of paper toweling.

3. Add 1 ml of resuspension solution to the bacterial pellet. Vortex or pipet up and down until the pellet is completely resuspended.

   Note: If processing ≤ 30 OD•ml of bacterial culture, reduce the volume of filtration lysis solution used to 0.6 ml.

4. Add 1 ml of filtration lysis solution and mix by inverting the tube briskly 6–8 times. DO NOT VORTEX. The solution will become viscous.

   Note: If processing ≤ 30 OD•ml of bacterial culture, reduce the volume of filtration lysis solution used to 0.6 ml.

   Note: The neutralization solution should be added to the lysate within 5 min of cell lysis.

5. Add 2 ml of neutralization solution and mix by inverting the tube briskly 6–8 times. DO NOT VORTEX. Allow the lysate to stand for 5 min before proceeding to the next step. The bacterial chromosomal DNA and cellular debris will form a visible white precipitate.

   Note: If processing ≤ 30 OD•ml of bacterial culture, reduce the volume of neutralization solution used to 1.2 ml.

6. Place a yellow lysate filtration midi column into a clean 15 ml conical low-speed centrifuge tube. Pour the entire neutralized lysate from step 5 into the filtration column.

7. Transfer the column and tube to a low-speed centrifuge with a swinging bucket rotor. Centrifuge for 5 min at 1,000 x g. During this step, place a green plasmid midi column into a clean 15 ml centrifuge tube.

   Note: In the event that residual lysate remains above the membrane stack, increase to 2,500 x g and centrifuge for an additional 5 min.

8. After centrifugation, discard the lysate filtration midi column and carefully pour the filtrate into the green plasmid midi column.

9. Centrifuge the plasmid midi column for 5 min at 1,000 x g. After the centrifugation is complete, check that all of the lysate has passed through the membrane stack.

   Note: In the event that residual lysate remains above the membrane stack, increase to 2,500 x g and centrifuge for an additional 5 min.

10. The wash solution is supplied as a 5x concentrate. Add 4 volumes (100 ml) of 95–100% ethanol or reagent-grade (denatured) ethanol before initial use.

11. Remove the plasmid midi column from the centrifuge tube and discard the filtrate. Replace the column into the same centrifuge tube and add 5 ml of wash solution. Centrifuge for 5 min at 1,000 x g.

12. Discard the wash solution and replace the plasmid midi column into the same centrifuge tube. Centrifuge for 5 min at 2,500 x g to expel residual wash solution.

    Note: Do not centrifuge for more than 5 min or excessive drying of the membranes may occur.

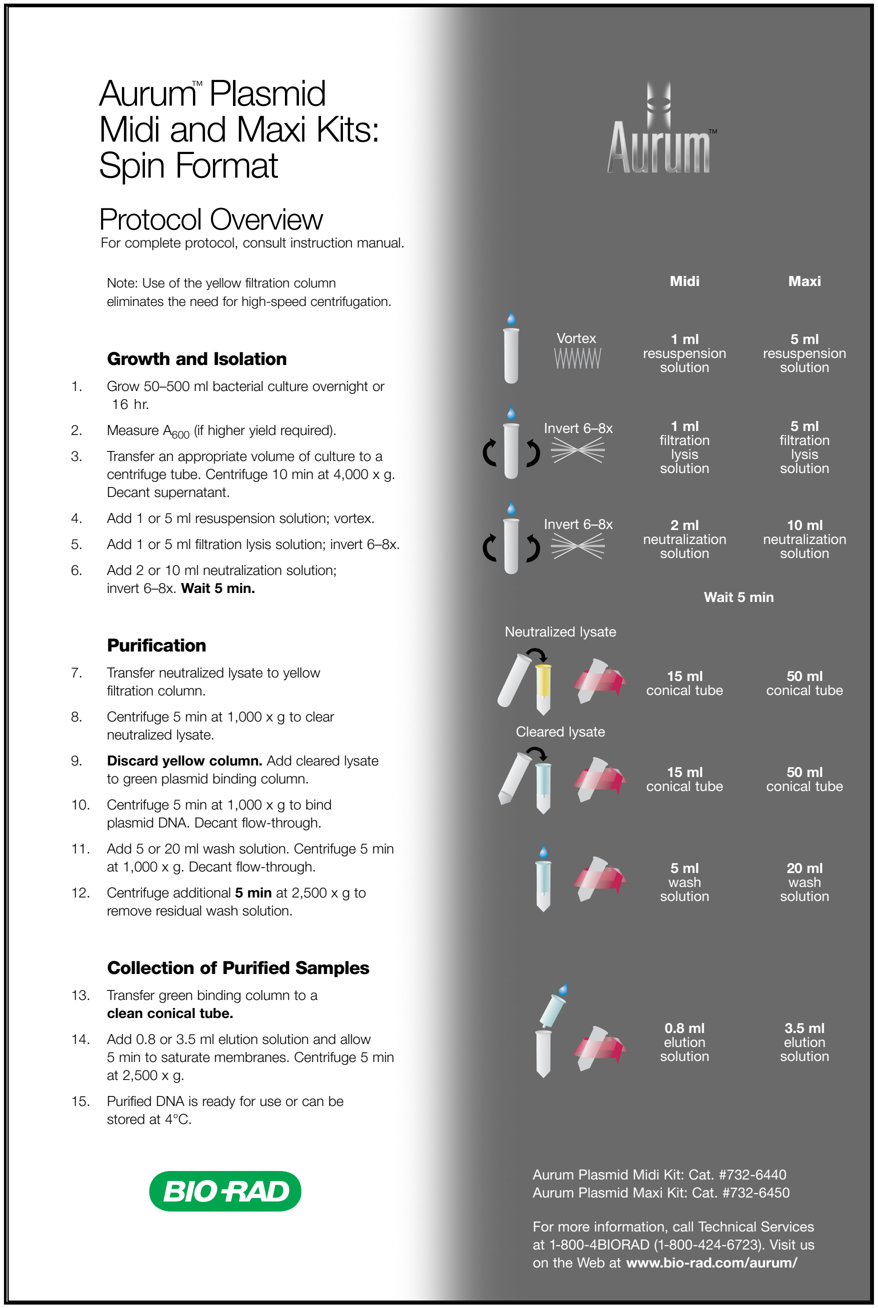
13. Remove the column from the centrifuge tube and wipe off any residual wash solution from the end of the column. Place the column into a clean 15 ml tube. Add 0.8 ml of elution solution directly onto the membranes at the base of the column and allow 5 min for the solution to saturate the membranes.

    Note: To obtain more concentrated plasmid DNA samples, the volume of elution solution may be reduced to 0.6 ml. Eluting with less than 0.6 ml may result in reduced plasmid yield.

14. Elute the plasmid DNA by centrifuging for 5 min at 2,500 x g.

    The eluted plasmid samples in the 15 ml tubes can be used immediately in mammalian cell transfections or in any other application. Alternatively, the plasmid can be stored at 4ºC for later use.

    A protocol overview is available (see Figure 6).



Aurum™ Plasmid Midi and Maxi Kits: Spin Format

Protocol Overview

For complete protocol, consult instruction manual.

Note: Use of the yellow filtration column eliminates the need for high-speed centrifugation.

Growth and Isolation

1. Grow 50–500 ml bacterial culture overnight or 16 hr.
2. Measure A_{600} (if higher yield required).
3. Transfer an appropriate volume of culture to a centrifuge tube. Centrifuge 10 min at 4,000 x g. Decant supernatant.
4. Add 1 or 5 ml resuspension solution; vortex.
5. Add 1 or 5 ml filtration lysis solution; invert 6–8x.
6. Add 2 or 10 ml neutralization solution; invert 6–8x. **Wait 5 min.**

Purification

7. Transfer neutralized lysate to yellow filtration column.
8. Centrifuge 5 min at 1,000 x g to clear neutralized lysate.
9. **Discard yellow column.** Add cleared lysate to green plasmid binding column.
10. Centrifuge 5 min at 1,000 x g to bind plasmid DNA. Decant flow-through.
11. Add 5 or 20 ml wash solution. Centrifuge 5 min at 1,000 x g. Decant flow-through.
12. Centrifuge additional **5 min** at 2,500 x g to remove residual wash solution.

Collection of Purified Samples

13. Transfer green binding column to a **clean conical tube.**
14. Add 0.8 or 3.5 ml elution solution and allow 5 min to saturate membranes. Centrifuge 5 min at 2,500 x g.
15. Purified DNA is ready for use or can be stored at 4°C.

Aurum Plasmid Midi Kit: Cat. #732-6440
Aurum Plasmid Maxi Kit: Cat. #732-6450

For more information, call Technical Services at 1-800-4BIORAD (1-800-424-6723). Visit us on the Web at **www.bio-rad.com/aurum/**

Fig. 6. Aurum plasmid midi protocol overview: spin format

Section 7 Troubleshooting Guide

Problem	Possible Cause	Possible Solution
Low plasmid yields/ low plasmid concentrations	Low copy number plasmid	Use high copy number plasmid. Alcohol-salt precipitation can be used to further concentrate if necessary.
	Poor plasmid propagation in culture	Inoculate large-scale cultures with overnight cultures generated from fresh colonies grown on a selective medium
		Determine optimum growth and plasmid propagation times for culture depending upon host, broth, etc.
		Check age and concentration of antibiotic
	Excessive amount of bacteria processed	Determine OD_{600} and reduce the OD•ml to ensure efficient lysis
	Incomplete resuspension of bacterial pellet	Check carefully to ensure that the bacteria pellet is completely resuspended prior to the addition of filtration lysis solution
	Excessive drying of column membranes prior to elution	See protocol. Do not exceed -25" Hg and indicated times in vacuum protocol.
	Insufficient ethanol added to wash solution concentrate	Check that correct amount was added

Problem	Possible Cause	Possible Solution
Low eluate volume (less than 0.5 ml)	Incomplete recovery of eluate from column	Centrifuge column for an additional 5 min at 2,500 x g
Bacterial chromosomal and DNA contamination	Excessive amount of bacteria processed	Determine OD_{600} of culture and do not exceed recommended OD•ml amounts
	Excessive agitation of lysate	Do not shake or vortex lysate after addition of lysis solution
RNA contamination	Excessive amount of bacteria processed	Determine OD_{600} and reduce OD•ml to ensure efficient lysis
	Reduced RNase activity due to age or storage conditions	Replace kit
Low $A_{260/280}$	Excessive amount of bacteria processed	Determine OD_{600} and reduce OD•ml processed
	Incomplete suspension of bacterial pellet	Ensure that the bacterial pellet is completely resuspended prior to addition of filtration lysis solution
	Incomplete mixing of resuspension, filtration lysis and neutralization solutions	Invert briskly 6–8 times after the addition of each reagent; check to ensure homogenous mixing
	Incomplete washing of column membranes	Use recommended wash solution volumes. Add correct amount of ethanol to wash solution concentrate.

Problem	Possible Cause	Possible Solution
More than one band on analytical gel	Presence of multimers – typical and variable depending upon plasmid, bacterial host, etc.	Try different host or growth conditions
	Nicking due to endonuclease activity	Use *endA*$^-$ host such as JM109, DH5α, or XL1-Blue
	Prolonged exposure to alkaline conditions	Add neutralization solution within 5 min after addition of filtration lysis solution
Plasmid nicked/ degraded	Use of *endA*$^+$ bacterial host	Use *endA*$^-$ host such as JM109, DH5α, or XL1-Blue
Plasmid performs poorly in enzymatic reactions	Contamination with residual salts, ethanol, or bacterial metabolites (may exhibit reduced $A_{260/280}$)	Use recommended culture volumes. Increase drying time by 1 min to remove wash solution.
	Excessive nicking due to *endA*$^+$ host	Use *endA*$^-$ host such as JM109, DH5α, or XL1-Blue
Plasmid performs poorly in mammalian cell transfections	Contamination with residual salts, ethanol, or bacterial metabolites (may exhibit reduced $A_{260/280}$)	Use recommended culture volumes. Increase drying time by 1 min to remove wash solution.
	Excessive nicking due to *endA*+ host	Use *endA*$^-$ host such as JM109, DH5α, or XL1-Blue

Problem	Possible Cause	Possible Solution
Column clogging	Excessive amount of bacteria processed	Do not exceed 50 OD•ml
	Reduced amount of bacteria processed	Reduce volumes of resuspension, filtration lysis and neutralization solutions to 0.6 ml, 0.6 ml and 1.2 ml, respectively, when processing ≤ 30 OD•ml of culture. Do not process less than 20 OD•ml.
	Incomplete mixing of of resuspension, lysis, and neutralization solutions	Invert briskly 6–8 times after the addition of each reagent; check to ensure homogeneous mixing
		Allow lysate to stand for 10 min after the addition neutralization solution before proceeding with filtration
	Insufficient negative pressure applied	Use a vacuum source capable of -20 to -23" Hg. Verify with vacuum regulator.
	Inadvertent use of other alkaline lysis reagent	Use only filtration lysis solution for bacterial lysis

Section 8
Ordering Information

Catalog #	Description
732-6460	**Aurum Plasmid 96 Kit,** 2 x 96 well preps, includes 2 grow blocks, 2 grow membranes, 2 binary units (2 lysate filtration plates and 2 plasmid binding plates), 2 collection plates, reagents, protocol overview, and instruction manual
732-6470	**Aurum Vacuum Manifold,** includes column adaptor plate, 4 replacement luer caps, A and B stages, waste collection tray, vacuum regulator and gauge, tubing, protocol overview, and instruction manual
732-6400	**Aurum Plasmid Mini Kit,** 100 preps, includes 100 plasmid binding columns, 100 capless collection tubes (2.0 ml), reagents, protocol overview, and instruction manual
732-6440	**Aurum Plasmid Midi Kit,** 20 preps, includes 20 lysate filtration columns, 20 plasmid binding columns, 20 adaptor caps, reagents, protocol overview, and instruction manual
732-6450	**Aurum Plasmid Maxi Kit,** 10 preps, includes 10 lysate filtration columns, 10 plasmid binding columns, 10 adaptor caps, reagents, protocol overview, and instruction manual